

## 32.6 Bayesian thread III: mixed models as hierarchical models

The Bayesian thread began in [Section 7.6](#) with a conjugate prior on  $(\boldsymbol{\beta}, \sigma^2)$  and continued in [Section 12.6](#) with credible sets and predictive distributions. Mixed models are where the two traditions meet most closely, because a random effect *is* a prior: the statement  $\mathbf{u} \sim N_q(\mathbf{0}, \mathbf{G})$  is formally indistinguishable from a normal prior on  $q$  unknown constants. What separates the methods of [Section 32.4](#) from a fully Bayesian analysis is only that they estimate  $\mathbf{G}$  and then treat it as known, which is empirical Bayes (Proposition 27.5.4), while a Bayesian puts a prior on  $\mathbf{G}$  too and integrates.

### 32.6.1 The hierarchical reading

Write the model in three levels:

$$\mathbf{Y} \mid \boldsymbol{\beta}, \mathbf{u}, \boldsymbol{\theta} \sim N_n(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}, \mathbf{R}), \quad \mathbf{u} \mid \boldsymbol{\theta} \sim N_q(\mathbf{0}, \mathbf{G}), \quad (\boldsymbol{\beta}, \boldsymbol{\theta}) \sim \pi. \quad (32.6.1)$$

Level one is a linear model with known coefficients; level two is the prior on  $\mathbf{u}$  that the experiment's sampling of levels supplies; level three is the part that is genuinely a matter of choice.

#### Proposition 32.6.1. BLUP as a posterior mean, REML as a marginal posterior

Consider [Eq. 32.6.1](#) with  $\mathbf{G}$  and  $\mathbf{R}$  positive definite and  $\text{rank}(\mathbf{X}) = p$ .

- (a) (*BLUP is a posterior mean.*) Fix  $\boldsymbol{\theta}$  and take the improper flat prior  $\pi(\boldsymbol{\beta}) \propto 1$ . Then the joint posterior of  $(\boldsymbol{\beta}, \mathbf{u})$  is

$$\begin{pmatrix} \boldsymbol{\beta} \\ \mathbf{u} \end{pmatrix} \mid \mathbf{y} \sim N_{p+q}\left(\begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{pmatrix}, \mathbf{C}^{-1}\right),$$

with  $\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}$  and  $\mathbf{C}$  as in Theorem 32.3.2. In particular  $E(\mathbf{u} \mid \mathbf{y}) = \hat{\mathbf{u}}$  is the BLUP,  $E(\boldsymbol{\beta} \mid \mathbf{y}) = \hat{\boldsymbol{\beta}}$  is the generalized least squares estimate, and  $\text{Cov}(\mathbf{u} \mid \mathbf{y})$  is the prediction error covariance of [Eq. 32.3.2](#).

- (b) (*REML is a marginal posterior.*) With the flat prior on  $\boldsymbol{\beta}$  and any prior  $\pi(\boldsymbol{\theta})$ , the marginal posterior of the variance components is

$$\pi(\boldsymbol{\theta} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) \exp\{\ell_R(\boldsymbol{\theta})\}, \quad (32.6.2)$$

with  $\ell_R$  the restricted log-likelihood of Definition 32.4.1. The REML estimate is therefore the mode of the marginal posterior under a flat prior on  $\boldsymbol{\theta}$ .

- (c) (*What the plug-in leaves out.*) Under (a) with  $\boldsymbol{\theta}$  replaced by an estimate, the reported posterior variance is  $\text{Cov}(\mathbf{u} \mid \mathbf{y}, \hat{\boldsymbol{\theta}})$ , whereas the correct marginal posterior variance is  $E\{\text{Cov}(\mathbf{u} \mid \mathbf{y}, \boldsymbol{\theta}) \mid \mathbf{y}\} + \text{Cov}\{E(\mathbf{u} \mid \mathbf{y}, \boldsymbol{\theta}) \mid \mathbf{y}\}$ . The plug-in omits the second term, so it is too small.

*Proof.*

- (a) With  $\pi(\boldsymbol{\beta}) \propto 1$  and  $\boldsymbol{\theta}$  fixed, the joint density of  $(\mathbf{y}, \mathbf{u})$  times the prior is proportional to  $\exp\{-\frac{1}{2}Q(\boldsymbol{\beta}, \mathbf{u})\}$  with  $Q$  the penalized criterion [Eq. 32.3.6](#), because  $-2 \log f(\mathbf{y} \mid \boldsymbol{\beta}, \mathbf{u}) =$

$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})^\top \mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})$  and  $-2\log f(\mathbf{u}) = \mathbf{u}^\top \mathbf{G}^{-1}\mathbf{u}$ , up to terms free of  $(\boldsymbol{\beta}, \mathbf{u})$ . By Theorem 32.3.2(b),  $Q$  is a quadratic in  $\mathbf{w} = (\boldsymbol{\beta}^\top, \mathbf{u}^\top)^\top$  with positive definite matrix  $\mathbf{C}$ , minimized at  $\hat{\mathbf{w}} = (\hat{\boldsymbol{\beta}}^\top, \hat{\mathbf{u}}^\top)^\top$ , so  $Q(\mathbf{w}) = (\mathbf{w} - \hat{\mathbf{w}})^\top \mathbf{C}(\mathbf{w} - \hat{\mathbf{w}}) + Q(\hat{\mathbf{w}})$ . The posterior is therefore  $N_{p+q}(\hat{\mathbf{w}}, \mathbf{C}^{-1})$ , and Theorem 32.3.2(c) identifies the blocks.

- (b) The joint posterior is  $\pi(\boldsymbol{\beta}, \boldsymbol{\theta} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) \exp\{\ell(\boldsymbol{\beta}, \boldsymbol{\theta})\}$  with  $\ell$  the marginal log-likelihood Eq. 32.4.1, the random effects having already been integrated out in Eq. 32.2.2. Completing the square as in Lemma 32.4.1(b),

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{P}\mathbf{y} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}),$$

so integrating over  $\boldsymbol{\beta} \in \mathbb{R}^p$  contributes the factor  $(2\pi)^{p/2} \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1/2}$  and

$$\int e^{\ell(\boldsymbol{\beta}, \boldsymbol{\theta})} d\boldsymbol{\beta} = \exp\left[-\frac{1}{2}\{(n-p)\log 2\pi + \log \det \mathbf{V} + \log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) + \mathbf{y}^\top \mathbf{P}\mathbf{y}\}\right].$$

By Eq. 32.4.5 the exponent is  $\ell_R(\boldsymbol{\theta})$  up to the additive constant  $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{X})$ , which does not involve  $\boldsymbol{\theta}$ . That is Eq. 32.6.2.

- (c) This is the decomposition of a variance into the mean of the conditional variance plus the variance of the conditional mean (Proposition 2.5.4), applied to the posterior distribution with  $\boldsymbol{\theta}$  as the conditioning variable. The second term is nonnegative definite. ■

Part (b) is Harville's (1974) observation, and it settles a question that the frequentist derivation leaves slightly mysterious. REML looked like a technical device for recovering lost degrees of freedom; it is in fact the exact marginal likelihood of the variance components once the fixed effects have been integrated out with a flat prior. The " $n-p$ " is not a correction but a consequence.

*Idea.* Three readings of the same computation. Frequentist:  $\hat{\mathbf{u}}$  minimizes prediction mean squared error among unbiased linear predictors. Penalized: it minimizes a sum of squares with a ridge penalty on  $\mathbf{u}$ . Bayesian: it is the posterior mean of  $\mathbf{u}$  under a normal prior with covariance  $\mathbf{G}$ . Henderson's equations compute all three at once, and the three readings differ only in what they say about  $\mathbf{G}$ : a covariance to be estimated, a penalty to be tuned, or a prior to be chosen.

## 32.6.2 Priors for variance components

The conditional distributions in Eq. 32.6.1 make the conjugate choices obvious. Given  $\mathbf{u}$  and  $\boldsymbol{\beta}$ , the sum of squares  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}\|^2$  is an ordinary normal likelihood for  $\sigma^2$ , so an inverse gamma prior is conjugate; given  $\mathbf{u}$ , the sum  $\|\mathbf{u}\|^2$  plays the same role for  $\sigma_a^2$ . This is what makes the Gibbs sampler below so short. What is not obvious is which inverse gamma.

*Warning.* The prior  $\sigma_a^2 \sim \text{IG}(\epsilon, \epsilon)$  with  $\epsilon$  small is often described as non-informative. It is not, and the limit does not exist. The marginal (restricted) likelihood  $\exp\{\ell_R\}$  is continuous and *strictly positive* at  $\sigma_a^2 = 0$ , because  $\sigma_a^2 = 0$  is a perfectly good model. Any prior density whose integral diverges at the origin, such as  $\pi(\sigma_a^2) \propto 1/\sigma_a^2$  — equivalently a flat prior on

$\log \sigma_a$  — therefore gives an improper posterior:  $\int \pi(\sigma_a^2) e^{\ell_R} d\sigma_a^2 = \infty$ . For small  $\epsilon$  the  $\text{IG}(\epsilon, \epsilon)$  posterior is proper but arbitrarily close to improper, and with few groups the answer depends on  $\epsilon$ . Gelman (2006) made this point and recommended instead a prior that is flat or half- $t$  on the *standard deviation*  $\sigma_a$ , which has finite mass near zero.

A convenient weakly informative default is the half-Cauchy on  $\sigma_a$  with scale  $A$ , chosen large relative to the plausible size of a group effect. It keeps the Gibbs sampler conjugate through the scale-mixture representation

$$\sigma_a^2 \mid \xi \sim \text{IG}(\tfrac{1}{2}, 1/\xi), \quad \xi \sim \text{IG}(\tfrac{1}{2}, 1/A^2),$$

whose marginal for  $\sigma_a$  is half-Cauchy(0,  $A$ ). The auxiliary variable  $\xi$  is sampled alongside everything else.

### 32.6.3 A Gibbs sampler in a dozen lines

For the one-way model with a flat prior on  $\mu$ , the four full conditionals are all standard. Given  $\mu, \sigma^2, \sigma_a^2$ , the effects are independent with

$$a_k \mid \text{rest} \sim N\left(\frac{(m/\sigma^2)(\bar{y}_k - \mu)}{m/\sigma^2 + 1/\sigma_a^2}, \frac{1}{m/\sigma^2 + 1/\sigma_a^2}\right),$$

which is Eq. 32.3.3 with the current parameter values; given the effects,  $\mu \sim N(\overline{y} - \overline{\mathbf{Z}\mathbf{a}}, \sigma^2/n)$ ; and the two variances have inverse gamma conditionals with shapes  $n/2$  and  $g/2$  plus their prior shapes. Nothing needs tuning.

```
import numpy as np

g, m = 12, 4
n = g * m
mu, sigma_a, sigma = 2.50, 0.20, 0.25          # the values used to simulate

rng = np.random.default_rng(20320048)
lab_effect = sigma_a * rng.standard_normal(g)
y = mu + np.repeat(lab_effect, m) + sigma * rng.standard_normal(n)

Y = y.reshape(g, m)                          # the REML fit of Section 32.4
ms_between = m * np.sum((Y.mean(axis=1) - Y.mean()) ** 2) / (g - 1)
ms_within = np.sum((Y - Y.mean(axis=1)[: , None]) ** 2) / (n - g)
reml_s2, reml_s2a = ms_within, (ms_between - ms_within) / m

def gibbs(y, g, m, draws=40000, burn=2000, prior="inverse-gamma", A=1.0, seed=0):
    """Posterior draws of (mu, sigma^2, sigma_a^2, a) with a flat prior on mu."""
    rng = np.random.default_rng(seed)
    Y = y.reshape(g, m)
    n = g * m
    mu_, s2, s2a, xi = Y.mean(), Y.var(), 0.1, 1.0
    out = np.empty((draws, 3 + g))
    for t in range(draws + burn):
        prec = m / s2 + 1 / s2a                # a_k | rest
        a = rng.normal((m / s2) * (Y.mean(axis=1) - mu_) / prec, np.sqrt(1 / prec))
        mu_ = rng.normal((y - np.repeat(a, m)).mean(), np.sqrt(s2 / n))    # mu |
        rest
        resid = y - mu_ - np.repeat(a, m)
        s2 = 1 / rng.gamma(0.001 + n / 2, 1 / (0.001 + resid @ resid / 2))
        if prior == "inverse-gamma":          # sigma_a^2 | rest
```

```

s2a = 1 / rng.gamma(0.001 + g / 2, 1 / (0.001 + a @ a / 2))
else:
    s2a = 1 / rng.gamma((g + 1) / 2, 1 / (1 / xi + a @ a / 2))
    xi = 1 / rng.gamma(1.0, 1 / (1 / A**2 + 1 / s2a))
if t >= burn:
    out[t - burn] = np.concatenate([[mu_, s2, s2a], a])
return out

```

[Run this code online](#)

```

# a shorter run than the 40 000 draws quoted in the text, so the cell finishes
quickly
short = gibbs(y, g, m, draws=4000, burn=500, prior="half-Cauchy", seed=12)
print(f"posterior median of sigma_a^2 {np.median(short[:, 2]):.4f}")
print(f"posterior mean of mu {short[:, 0].mean():.4f}")
print(f"REML estimate of sigma_a^2 {reml_s2a:.4f}")

```

[Run this code online](#)

### Example. The proficiency study, four ways

Run the sampler for 40 000 draws on the data of Example (A laboratory proficiency study). Under the  $IG(0.001, 0.001)$  prior the posterior median of  $\sigma_a^2$  is 0.0356; under the half-Cauchy prior with  $A = 1$  it is 0.0438. The REML estimate 0.0386 sits between them. The two priors therefore disagree by about a fifth of the estimate, with twelve laboratories — which is already a reason to report the prior.

The posterior means of the twelve effects track the BLUPs closely: their correlation exceeds 0.99 and the largest in absolute value is 0.294 against the BLUP's 0.321, slightly more shrunk — not because the posterior sits below the REML estimate, since its mean 0.0431 is above it, but because the shrinkage factor  $B = m\sigma_a^2/(\sigma^2 + m\sigma_a^2)$  is concave in  $\sigma_a^2$ : averaging  $B$  over a posterior with mass near zero weights the data less than  $\hat{\sigma}_a^2$  does. Their posterior standard deviations average 0.1218, against the plug-in prediction standard error 0.1156 of Eq. 32.3.4: 5 per cent wider, which is the uncertainty about  $\sigma_a^2$  that the plug-in throws away, exactly as [Proposition 32.6.1\(c\)](#) says. The same happens for the fixed effect: the posterior standard deviation of  $\mu$  is 0.0709, against the plug-in  $\sqrt{\hat{\lambda}_1/n} = 0.0672$ .

Panel (b) of [Figure 32.6.1](#) repeats the analysis on the first 5 laboratories. The posterior medians are now 0.0435 and 0.0693, a difference of sixty per cent, against a REML estimate of 0.0493. With five groups the prior is doing a large part of the work, and saying so is not optional.

## 32.6.4 What the Bayesian version buys, and what it costs

Three gains. The uncertainty about  $\theta$  is carried through automatically, so intervals for fixed effects and prediction intervals for random effects are honestly wider, with no need for the corrections of [Section 32.5](#). The boundary problem disappears: the posterior of  $\sigma_a^2$  is a proper distribution on  $[0, \infty)$ , with no point mass and no chi-bar-squared mixture to calibrate. And the same machinery handles models with no closed form, which is what [Chapter 33](#) needs for richer covariance structures and [Chapter 40](#) needs once the response is not normal.

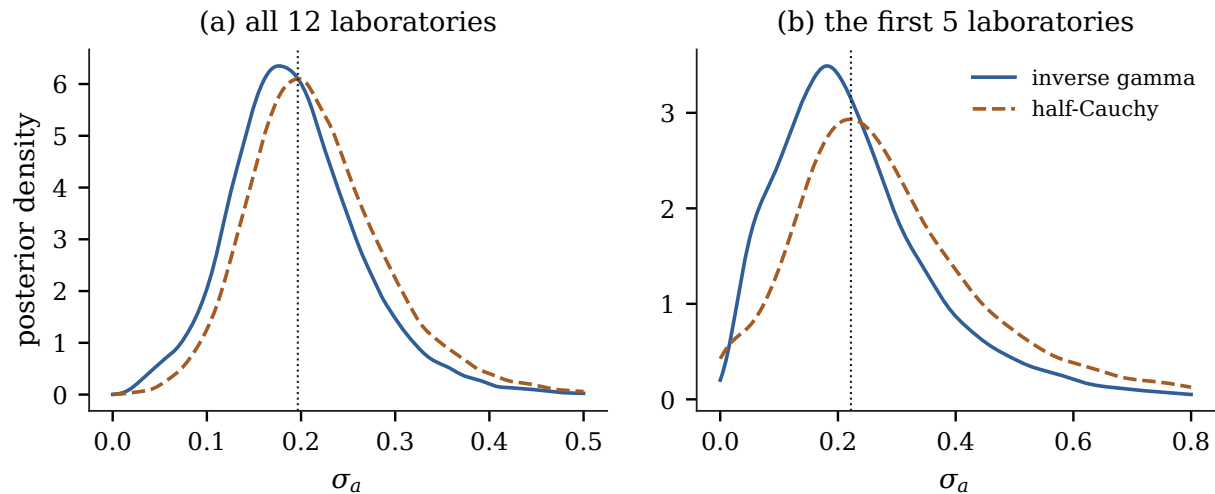


Figure 32.6.1: Posterior densities of the between-laboratory standard deviation  $\sigma_a$  under two priors, from a Gibbs sampler with 40 000 draws. The dotted line marks the REML estimate. (a) All twelve laboratories. (b) The first five, where the prior matters a great deal.

Two costs. A prior for the variance components must be chosen, and with few groups it matters, as panel (b) shows; the only defensible practice is to state it and show the sensitivity. And the answers are not exactly the frequentist ones: the two frameworks agree on the point estimates, but they part company on the spread, and the frequentist coverage of a credible interval for a variance component is not guaranteed (Proposition 12.6.4).

## 32.6.5 Exercises

### 32.6.5.1 A. Check your understanding

#### Exercise. A1

Explain in two sentences why a random effect and a prior are formally the same object, and name one thing that distinguishes them in practice.

### 32.6.5.2 B. Practice

#### Exercise. B1

Show that in the one-way model the marginal posterior density of  $\sigma_a^2$  under the prior  $\pi(\sigma_a^2) \propto \sigma_a^{-2}$  is not integrable, by showing that  $\exp\{\ell_R(\sigma_a^2)\}$  tends to a strictly positive limit as  $\sigma_a^2 \downarrow 0$  with  $\sigma^2$  profiled out. Contrast with the behaviour of the prior  $\pi(\sigma_a) \propto 1$ .

*Solution.* By Eq. 32.4.8,  $-2\ell_R$  is continuous in  $(\lambda_1, \lambda_2)$  on  $\lambda_1 \geq \lambda_2 > 0$ , and at  $\sigma_a^2 = 0$  it equals  $(n-1)\log \lambda_2 + (\text{SSB} + \text{SSE})/\lambda_2$ , finite and minimized at a positive  $\lambda_2$ . So  $\exp\{\ell_R\} \rightarrow c > 0$  as  $\sigma_a^2 \downarrow 0$ , and  $\int_0^\delta c \sigma_a^{-2} d\sigma_a^2 = \infty$ . With  $\pi(\sigma_a) \propto 1$  the Jacobian gives  $\pi(\sigma_a^2) \propto \sigma_a^{-1}$ , and  $\int_0^\delta \sigma_a^{-1} d(\sigma_a^2) = 2 \int_0^{\sqrt{\delta}} d\sigma_a < \infty$ : the posterior is proper near the origin.

**Exercise. B2**

In Proposition 32.6.1(c), suppose  $\theta$  takes only two values with posterior probabilities  $1/2$  each, giving conditional posterior means  $\hat{u}_1, \hat{u}_2$  and common conditional variance  $v$  for a coordinate. Compute the correct marginal posterior variance, and say how large the disagreement between  $\hat{u}_1$  and  $\hat{u}_2$  must be before the plug-in understates the standard deviation by ten per cent.

**32.6.5.3 C. Going deeper**

**Exercise. C1**

Formula Eq. 32.6.2 shows REML is the posterior mode of  $\theta$  under a flat prior on  $\theta$ . A mode is not invariant to reparameterization: the mode of the posterior of  $\sigma_a^2$  is not the square of the mode of the posterior of  $\sigma_a$ . Reconcile this with Theorem 32.4.2(c), which says the REML estimate is well defined. What is invariant, and what is not?

**Exercise. C2**

Take the balanced one-way model with  $\sigma^2$  known,  $m = 1$  so that  $\bar{y}_k = y_k$ , and  $\mu$  known to be zero. Show that plugging the unbiased moment estimate of  $\sigma_a^2$  into Eq. 32.3.3 gives a shrinkage factor of the form  $1 - c\sigma^2 / \sum_k y_k^2$ , and compare with the James–Stein estimator of Theorem 27.5.2. Which constant  $c$  does the plug-in give, and which does James–Stein use?

*Solution.* With  $m = 1$  and  $\mu = 0$  the data are  $y_k \sim N(a_k, \sigma^2)$  with  $a_k \sim N(0, \sigma_a^2)$ , so marginally  $y_k \sim N(0, \sigma^2 + \sigma_a^2)$ . The moment estimate from  $\sum_k y_k^2$  is  $\hat{\sigma}_a^2 = \sum_k y_k^2 / g - \sigma^2$ , and

$$B = \frac{\hat{\sigma}_a^2}{\hat{\sigma}_a^2 + \sigma^2} = 1 - \frac{g\sigma^2}{\sum_k y_k^2},$$

so  $c = g$  in the stated form. The James–Stein estimator of Theorem 27.5.2 uses  $1 - (g - 2)\sigma^2 / \sum_k y_k^2$ . The difference is the same degrees-of-freedom correction that separates ML from REML:  $g - 2$  is the constant that makes  $(g - 2)\sigma^2 / \sum y_k^2$  unbiased for  $\sigma^2 / (\sigma^2 + \sigma_a^2)$ , and it is what makes the estimator dominate the unshrunk one for every  $\alpha$ , not merely on average.